



Exam

Final 2024

Question 1

1. Compare briefly between Systems that "Act / Behave Rationally", and Systems that "Act / Behave Humanly".
2. Describe briefly Intelligent Agents.
3. Discuss PEAS, and how is it employed to specify a task environment.

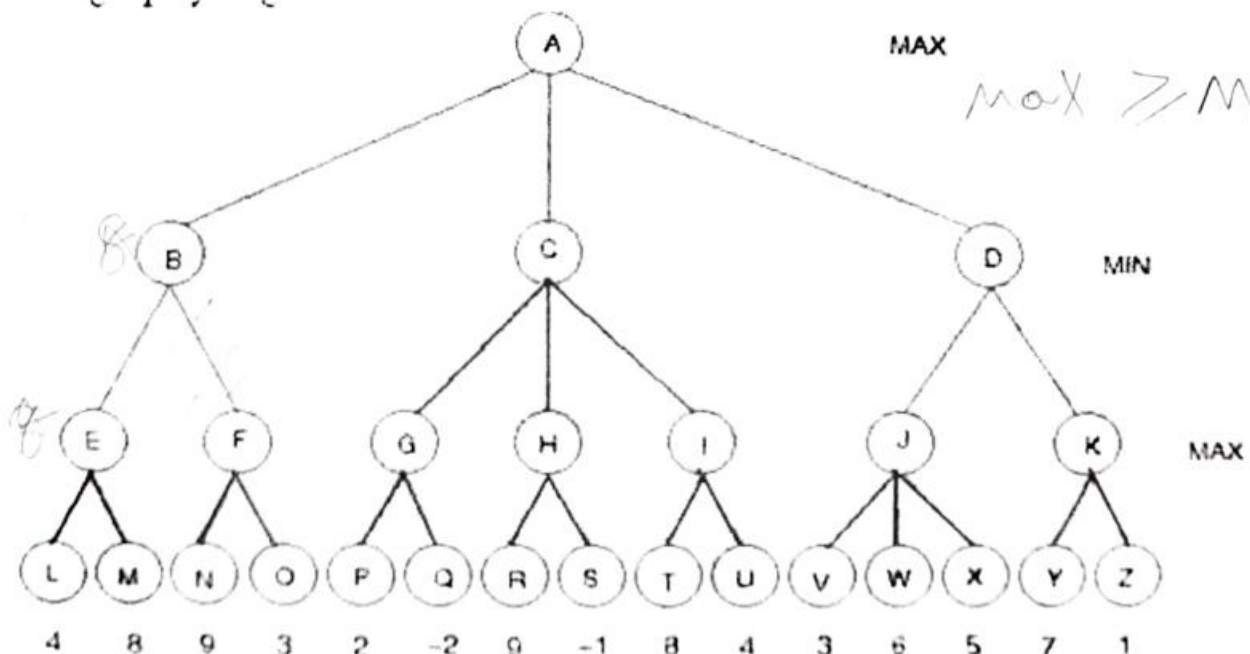
Question 3

Apply the k-means algorithm to cluster the following patterns into 3 partitions using the Manhattan distance measure based on selecting P_1 , P_6 and P_7 as initial centroids. Show the first iteration of the clustering process and compute the new centroids for the next iteration.

	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9
x	1	5	9	8	2	5	5	7	3
y	1	7	1	1	1	6	5	1	1
	1	2	3	3	1	2	2	3	1

Question 4

1. Apply the Min-Max Procedure and then the Alpha-Beta Pruning Algorithm for the following 2 players game trees.

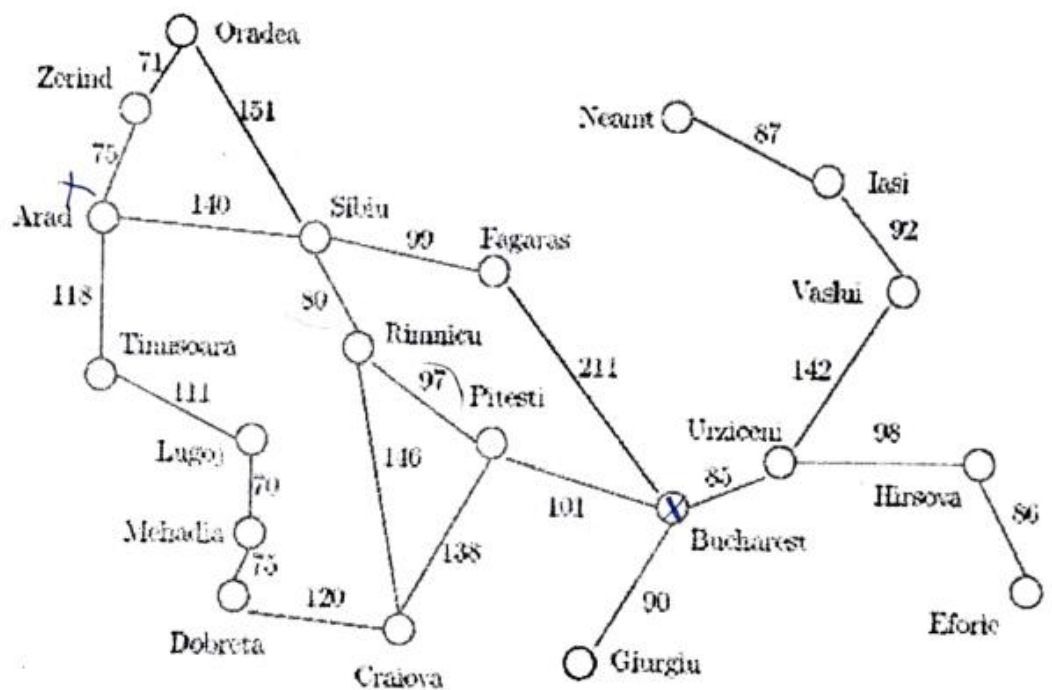


**Question 5**

2. Given the Arad/Bucharest route finding problem and the heuristics table below, show the list of visited nodes using the following search techniques:

- Breadth first
- Depth First
- Best first.
- A* Search

n	H(n)
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374





Exam

Midterm 2025

Question 1. True / False:

1. The game of Poker is a single agent. ☒
2. A solution to a problem is a path from the initial state to a goal state. Solution quality is measured by the path cost function, and an optimal solution has the highest path cost among all solutions. ☒
3. likes(jin, jax). the same as likes(jax, jin). ☒
4. food(b, c, d(a)). is ?-food(X, Y). a true query? ☒
5. if sky is blue, everyone likes it. Write this statement as likes(sky, everyone) :- blue(sky). ☒
6. ?- X is -(5,3,1). return an error. ☒
7. ?- 3+2 is 5. same as ?- 5 is 3+2. ☒

Question 2. MCQ

1. Satellite Image Analysis System is (Choose the one that is not applicable).
a) Episodic b) Semi-Static
c) Single agent ☒ d) Partially Observable
2. Categorize Crossword puzzle in Fully Observable / Partially Observable.
☒ a) Fully Observable b) partially Observable
c) All of the mentioned d) None of the mentioned
3. Which were built in such a way that humans had to supply the inputs and interpret the outputs?
☒ a) Agents b) AI system c) Sensor d) Actuators
4. What is state space?
a) The whole problem b) Your Definition to a problem
c) Problem you design ☒ d) Representing your problem with variable and parameter
5. What is the main task of a problem-solving agent?
a) Solve the given problem and reach to goal
b) To find out which sequence of action will get it to the goal state
☒ c) All of the mentioned
d) None of the mentioned
6. What is the full form of "AI"?
☒ a) Artificially Intelligent
☒ b) Artificial Intelligence
c) Artificially Intelligence
d) Advanced Intelligence ☒
7. Which of the following is/are Uninformed Search technique/techniques?
a) Breadth First Search (BFS) b) Depth First Search (DFS)
c) Bidirectional Search ☒ d) All of the mentioned
8. What is Artificial Intelligence?
a) Artificial Intelligence is a field that aims to make humans more intelligent ☒
b) Artificial Intelligence is a field that aims to improve the security ☒
☒ c) Artificial Intelligence is a field that aims to develop intelligent machines
d) Artificial Intelligence is a field that aims to mine the data

~~Artificial Intelligence~~

9. Who is the inventor of Artificial Intelligence?
- a) Geoffrey Hinton b) Andrew Ng
~~c) John McCarthy~~ d) Jürgen Schmidhuber
10. Which search implements stack operation for searching the states?
- a) Depth-limited search ~~b) Depth-first search~~ Q → B
c) Breadth-first search d) None of the mentioned S → D
11. Which of the following is the branch of Artificial Intelligence?
- ~~a) Machine Learning~~ b) Cyber forensics
c) Full-Stack Developer d) Network Design
12. What is the goal of Artificial Intelligence?
- a) To solve artificial problems
b) To extract scientific causes
c) To explain various sorts of intelligence
~~d) To solve real-world problems~~
13. Which of the following is an application of Artificial Intelligence?
- a) It helps to exploit vulnerabilities to secure the firm
~~b) Language understanding and problem-solving (Text analytics and NLP)~~
c) Easy to create a website ✓
d) It helps to deploy applications on the cloud ✓
14. Breadth-first search always expands the _____ node in the current fringe of the search tree.
- ~~a) Shallowest~~ b) Child node
c) Deepest d) Minimum cost
15. Which of the following is a component of Artificial Intelligence?
- ~~a) Learning~~ b) Training
c) Designing d) Puzzling
16. Prolog is a programming language based on which paradigm?
- a) Imperative programming b) Object-oriented programming
c) Functional programming ~~d) Logic programming~~
17. In Prolog, knowledge is represented using:
- a) Variables b) Loops
~~c) Facts and rules~~ d) Functions
18. What is the correct syntax for a Prolog rule representing "X is the mother of Y"?
- ~~a) mother(X, Y) :-~~ b) X(mother, Y) :- ✓
c) Y is mother(X) :- d) mother(X) = Y :- ✓ ? - mother (✓)
19. likes(joe, fish).
likes(mary, fish).
- likes(joe, mary).
likes(albert, joe).



20. What is the function of an Artificial Intelligence "Agent"?

- a) Mapping of goal sequence to an action
- b) Work without the direct interference of the people
- ☒ c) Mapping of precept sequence to an action
- d) Mapping of environment sequence to an action

21. Depth-first search always expands the _____ node in the current fringe of the search tree.

- a) Shallowest
- b) Child node
- ☒ c) Deepest
- d) Minimum cost

22. Which of the following machine requires input from the humans but can interpret the outputs themselves?

- a) Actuators
- b) Sensor
- ☒ c) Agents
- d) AI system

23. Which of the following can improve the performance of an AI agent?

- a) Perceiving
- b) Learning
- c) Observing
- ☒ d) All of the mentioned

24. Which of the following is/are the composition for AI agents?

- a) Program only
- b) Architecture only
- ☒ c) Both Program and Architecture
- d) None of the mentioned

25. On which of the following approach A basic line following robot is based?

- ☒ a) Applied approach
- b) Weak approach
- c) Strong approach
- d) Cognitive approach

26. Which of the following is an example of artificial intelligent agent/agents?

- a) Autonomous Spacecraft
- b) Human
- ☒ c) Robot
- d) All of the mentioned

27. What is an AI 'agent'?

- a) Takes input from the surroundings and uses its intelligence and performs the desired operations
- b) An embedded program controlling line following robot
- c) Perceives its environment through sensors and acting upon that environment through actuators
- ☒ d) All of the mentioned

28. Which of the following environment is strategic?

- a) Rational
- ☒ b) Deterministic
- c) Partial
- d) Stochastic

29. In Prolog, which operator is used to represent logical OR?

- a) &&
- b) ,
- c) !
- ☒ d) ;

30. In Prolog, which symbol is used to represent the end of a query?

- a) !
- ☒ b) .
- c) :
- d) ?

31. Which search strategy is also called as blind search?

- ☒ a) Uninformed search
- b) Informed search
- c) Simple reflex search
- d) All of the mentioned

32. Which search is implemented with an empty first-in-first-out queue?

- ☒ a) Depth-first search
 - b) Breadth-first search
 - c) Bidirectional search
 - d) None of the mentioned
33. Which search algorithm imposes a fixed depth limit on nodes?
- ☒ a) Depth-limited search
 - b) Depth-first search
 - c) Iterative deepening search
 - d) Bidirectional search